

José Henrique Targino Dias Gois

Software Developer | Rust, Linux and Low-Level Systems

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Profile

Software developer focused on Rust, Linux interfaces, userspace environments, and low-level systems. Builds open-source abstractions and experimental runtimes involving system calls, ELF binaries, assembly bootstrap, and `no_std` Rust. PhD-trained researcher with extensive experience in reproducible computational work and quantitative problem solving.

Technical Skills

Programming	Rust, C, C++, Python, Shell, MATLAB
Systems	Linux, system calls, ELF, assembly bootstrap, <code>no_std</code> , userspace runtimes
Tools	Git, GitHub Actions, Cargo, QEMU, GDB, scientific computing and data analysis

Selected Projects

Userspace <i>Experimental userspace environment</i>	Rust / Assembly crates.io/crates/userspace
• Bootstraps Rust from assembly and provides low-level runtime facilities for Linux userspace experimentation. • Implements and organizes interfaces for system calls, memory management, file access, processes, and ELF-oriented development.	
Ample <i>Linux kernel interface abstractions</i>	Rust crates.io/crates/ample
• Develops reusable Rust abstractions over interfaces exposed by the Linux kernel. • Supports low-level and <code>no_std</code>-oriented software while keeping kernel-facing behavior explicit.	

Experience

Open-Source Systems Projects <i>Independent Rust and Linux development</i>	Current Natal, Brazil
Develops and maintains low-level Rust libraries and experimental userspace software, including build automation, cross-target compilation, and repository-based CI workflows.	
Graduate Researcher <i>Instituto do Cérebro, Universidade Federal do Rio Grande do Norte</i>	2012–2020 Natal, Brazil
• Designed reproducible computational workflows for electrophysiology, microscopy, neuronal activity, and behavioral datasets. • Published peer-reviewed research across experimental neuroscience, computational analysis, and bibliometrics.	
Programmer <i>AR Consultoria em Informática</i>	2005–2006 Natal, Brazil
Maintained and developed enterprise information-management modules using Delphi/Pascal and Firebird SQL.	

Education

BSc in Computer Science, UFRN	Expected 2028
BSc in Information Security, UNINTER	Expected 2029
BSc in Information Technology, UFRN	2025
PhD in Neuroscience, UFRN — Computational neuroscience	2020
MSc in Neuroscience, UFRN — Molecular neuroscience	2015
BSc in Science and Technology, UFRN	2012

Additional Information

Portuguese (native), English (professional working proficiency), Spanish (intermediate). Co-author of peer-reviewed research published in *Cell Reports*, *Journal of Neurophysiology*, *Frontiers in Human Neuroscience*, and *PLOS ONE*.